

FREEDOM INTERNATIONAL SCHOOL

WORKSHEET- MCQ (SOLUTIONS)

PHYSICS

OSCILLATIONS

CLASS XI

1. A particle has displacement y given by $y = 3 \sin(5\pi t + \pi)$, where y is in metre and t is in second. What are frequency and period of motion?

(a) 0.4 Hz, 2.5 s

(b) 2.5 Hz, 0.4 s

(c) 2.5 Hz, 2.5 s

(d) 0.4 Hz, 0.4 s

Ans: (b) 2.5 Hz, 0.4 s

The standard equation is $y = a \sin(\omega t + \phi) \dots (i)$

where a is amplitude, ω the angular velocity and ϕ the phase.

Given, equation is $y = 3 \sin(5\pi t + \pi) \dots (ii)$

Comparing Eqs. (i) and (ii),

we get $\omega = 5\pi$

Also, $\omega = 2\pi \nu$

$$\nu = \frac{\omega}{2\pi} = \frac{5\pi}{2\pi} = 2.5 \text{ Hz}$$

$$T = \frac{1}{\nu} = \frac{1}{2.5} = 0.4 \text{ s}$$

2. A particle executing simple harmonic motion has a time period of 4 s. If the particle starts moving from the mean position at $t=0$, then after how much interval of time from $t=0$, will its displacement be half of its amplitude?

(a) $\frac{1}{3}$ s

(b) $\frac{1}{2}$ s

(c) $\frac{2}{3}$ s

(d) $\frac{1}{6}$ s

Ans: (a) $\frac{1}{3}$ s

$$T = 4\text{sec}$$

Displacement equation is S.H.M is given by

$$y = a \sin \omega t \dots (1)$$

$$\text{If } y = \frac{a}{2}$$

Equation (1) becomes,

$$\frac{a}{2} = a \sin \omega t$$

$$\sin \omega t = 0.5$$

$$\omega t = \sin^{-1}(0.5) = \frac{\pi}{6}$$

$$\frac{2\pi}{T} t = \frac{\pi}{6}$$

$$t = \frac{T}{12}$$

$$t = \frac{4}{12} = 1/3 \text{ s}$$

3. A body is executing the S.H.M. with an angular frequency of 2 rad/s. Velocity of the body at 20 mm displacement, when amplitude of motion is 60 mm, is
 (a) 90 mm/s (b) 118 mm/s (c) 113 mm/s (d) 131 mm/s

Ans: (c) 113 mm/s

We know that, the particle velocity

$$v = \omega \sqrt{A^2 - y^2}$$

Given that $\omega = 2 \text{ rad/s}$,

$$A = 60 \text{ mm} = 0.06 \text{ m}$$

$$\text{and } y = 20 \text{ mm} = 0.02 \text{ m}$$

$$\therefore v = 2 \sqrt{(0.06)^2 - (0.02)^2}$$

$$= 0.113 \text{ m/s}$$

$$= 113 \text{ mm/s}$$

4. What is the maximum acceleration of the particle executing the S.H.M., $y = 2 \sin \left[\frac{\pi t}{2} + \Phi \right]$, where y is in cm?
 (c) $\frac{\pi}{2} \text{ cm/s}^2$ (b) $\frac{\pi^2}{2} \text{ cm/s}^2$ (c) $\frac{\pi}{4} \text{ cm/s}^2$ (d) $\frac{\pi^2}{4} \text{ cm/s}^2$

Ans: (b) $\frac{\pi^2}{2} \text{ cm/s}^2$

$$A = 2 \text{ cm}, \omega = \pi/2 \text{ rad/s}$$

$$a_{\max} = \omega^2 A = \frac{\pi^2}{2} \text{ cm/s}^2$$

5. The maximum velocity and maximum acceleration of a body moving in a simple harmonic motion are 2 m/s and 4 m/s² respectively. Then the angular velocity will be
 (a) 4 rad/s (b) 3 rad/s (c) 2 rad/s (d) 8 rad/s

Ans: (c) 2 rad/s

$$v_{\max} = \omega A = 2 \text{ m/s} \quad a_{\max} = \omega^2 A = 4 \text{ m/s}^2$$

$$\omega = a_{\max}/v_{\max} = 4/2 = 2 \text{ rad/s}$$

6. Average value of K.E. and P.E. over entire time period is

(a) $0, \frac{1}{2} m\omega^2 A^2$

(b) $\frac{1}{2} m\omega^2 A^2, 0$

(c) $\frac{1}{2} m\omega^2 A^2, \frac{1}{2} m\omega^2 A^2$

(d) $\frac{1}{4} m\omega^2 A^2, \frac{1}{4} m\omega^2 A^2$

Ans: (d) $\frac{1}{4} m\omega^2 A^2, \frac{1}{4} m\omega^2 A^2$

7. The spring constants of three springs connected to a mass 'm' are shown in the figure. When mass m oscillates, what are the effective spring constant and the time period of vibration?

(a) $4k, 2\pi\sqrt{\frac{m}{4k}}$

(b) $3k, 2\pi\sqrt{\frac{m}{3k}}$

(c) $2k, 2\pi\sqrt{\frac{m}{2k}}$

(d) $k, 2\pi\sqrt{\frac{m}{k}}$

Ans: (d) $k, 2\pi\sqrt{\frac{m}{k}}$

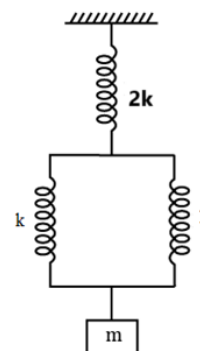
In lower part k and k parallel

$$k_{\text{net}} = 2k$$

2k and 2k are in series

$$k_{\text{net}} = k$$

$$T = 2\pi\sqrt{\frac{m}{k}}$$



8. A mass m is suspended from a spring. Its frequency of oscillation is f . The spring is cut into two halves and the same mass is suspended from one of the two pieces of the spring. The frequency of oscillation of mass will be

(a) $\sqrt{2}f$ (b) $f/2$ (c) f (d) $2f$

Ans: (a) $\sqrt{2}f$

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$k' = 2k$$

$$f' = \frac{1}{2\pi} \sqrt{\frac{k'}{m}} = \frac{1}{2\pi} \sqrt{\frac{2k}{m}} = \sqrt{2}f$$

9. A simple pendulum has a time period of 1 s. In order to increase the time period to 2 s,
- (a) the mass of the bob should be doubled
 (b) the length of the pendulum should be doubled
 (c) the length of the pendulum should be increased by a factor of 4
 (d) the length of the pendulum should be decreased by a factor of 4

Ans: (c) the length of the pendulum should be increased by a factor of 4

$$l_2/l_1 = T_2^2/T_1^2$$

$$l_2/l_1 = 4/1$$

10. On a planet, a freely falling body takes 2 s when it is dropped from a height of 8 m. The time period of a simple pendulum of length 1m on that planet is

(a) 1.42 s (b) 3.14 s (c) 2.92 s (d) 5.36 s

Ans: (b) 3.14 s

Height, $h = 8m$,

Time taken to reach the ground, $t = 2 \text{ sec}$

But for a body dropped, $t = \sqrt{\frac{2h}{g}}$

$$2 = \sqrt{\frac{16}{g}}$$

$$g = \frac{16}{4} = 4m/s^2 \text{ on that planet}$$

Time period of pendulum, $T = 2\pi\sqrt{\frac{l}{g}}$

$$T = 2\pi\sqrt{\frac{1}{4}} \\ = \pi \text{ s}$$

For questions 11 to 15, two statements are given- one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer to these questions from the options as given below.

- A. Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
- B. Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
- C. Assertion is true but Reason is false.
- D. Both Assertion and Reason are false.

11. **Assertion:** All oscillatory motions are necessarily periodic motion but all periodic motion are not oscillatory.

Reason: Simple pendulum is an example of oscillatory motion.

Ans: B

12. **Assertion:** The graph of total energy of a particle in SHM w.r.t., position is a straight line with zero slope.

Reason: Total energy of particle in SHM remains constant throughout its motion.

Ans: A

13. **Assertion:** The periodic time of a hard spring is less as compared to that of a soft spring.

Reason: The periodic time depends upon the spring constant, and spring constant is large for hard spring.

Ans: A

14. **Assertion:** In S.H.M., acceleration always acts towards mean position.

Reason: In extreme position of a particle executing S.H.M., both velocity and acceleration are zero.

Ans: C

15. **Assertion:** Simple harmonic motion is a uniform motion.

Reason: Simple harmonic motion is not the projection of uniform circular motion.

Ans: D